Department of Electrical Engineering
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Introduction to Prealgebra
Ch. 3 Number Theory

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OUTLINE

- Introduction
- Multiples
- Divisibility
- Prime Numbers
- Prime Factorization
- Least Common Multiple
- Divisors
- Greatest Common Divisor
- Summary

INTRODUCTION

- **Number theory is the study of integers**
  - Multiples, least common multiples (LCM)
  - Prime numbers
  - Divisors, greatest common divisors (GCD)

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MULTIPLES

- **Definition**
  - Let a and b be numbers.

> a is a multiple of b if there is an integer n such that $a = bn$.

  - Example
    - 21 is a multiple of 7 because $21 = 7 \cdot 3$.
    - 21 is NOT a multiple of 8.

MULTIPLES

- **Example**
  - Both 147 and 357 are multiples of 7. Is $147 + 357$ a multiple of 7?
  - Suppose k is a multiple of 7. Must $k + 7$ be a multiple of 7?
  - Suppose r and s are multiples of 7. Must $r + s$ be a multiple of 7?
  - Suppose k is a multiple of 7. Is $k + 23$ a multiple of 7?

MULTIPLES

- **Adding (or subtracting) multiples**

If a and b are multiples of c, then both $a + b$ and $a - b$ are multiples of c.

  - Why?

MULTIPLES

- **Counting with multiples**

If we start from a multiple of n and count either upward or downward by n, then all the number we hit will be multiples of n.

$$700, 693, 686, 679, 672, 665, 658, 651, \ldots$$

– Why?

MULTIPLES

- **Adding (or subtracting) non-multiples**

If a is a multiple c, but b is not a multiple of c. Then neither $a+b$ nor $a-b$ is a multiple of c.

– Why?

MULTIPLES

- **Example**
  - What number between 100 and 200 is both a perfect square and a multiple of 7?
    - Important: between 100 and 200 means 100 and 200 are NOT included.

MULTIPLES

- **Example**
  - What is the greatest 3-digit number that is a multiple of 13?

MULTIPLES

- **Example**
  - How many integers between 2 and 1004 are multiples of 5?
  - How many integers between 150 and 300 are multiples of 9?

MULTIPLES

- **Multiple of multiples**

> If a is a multiple b, then every multiple of a is also a multiple of b.

  - Why?

  - Example
    - Must every multiple of 15 also a multiple of 3?
    - Must every multiple of 3 also a multiple of 15?

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DIVISIBILITY

- **Definition**
  - Let *a* be a number, and let *b* be a nonzero number. We say that *a* is **divisible** by *b* if

$$a \div b \text{ is an integer.}$$

  - Example
    - 12 is divisible by 3.
    - 72 is divisible by 9.
    - 144 is divisible by 12.
  - If a and b are not both 0, then
    - "***a* is a multiple of *b***" is the same as "***a* is divisible by *b***".

DIVISIBILITY

- **Divisible by 10**

> Any integer that has 0 on its units digit is divisible by 10.

  - Why?

  - Example
    - 30, 70, 120, 13670, 900200, …

DIVISIBILITY

- **Divisible by 5**

> Any integer that has 0 or 5 on its units digit is divisible by 5.

  - Why?

  - Example
    - 35, 60, 105, 723765, ……

DIVISIBILITY

- **Divisible by 2**

> Any integer that has an even units digit (0, 2, 4, 6, 8) is divisible by 2.
>
> Any integer that has an odd units digit (1, 3, 5, 7, 9) is NOT divisible by 2.

  - Why?

  - Example
    - 62, 78, 976, 67892678

    - 13, 79, 81, 765, 683117

DIVISIBILITY

- **Example**
  - Which of the following numbers are divisible by 4?
    12, 312, 512, 2512, 4312
  - Is every multiple of 100 also a multiple of 4?
  - Is 5,687,623,688 divisible by 4?
  - Is 4,650,310 divisible by 4?

DIVISIBILITY

- **Divisible by 4**

A positive integer is divisible by 4 if the number formed by the last two digits of the original integer is divisible by 4.

- Why?

- Example
  - 51,378,988
  - 123,456,789

DIVISIBILITY

- **Example**
  - Is 765 divisible by 9?

$$\begin{aligned} 765 &= 7 \cdot 100 + 6 \cdot 10 + 5 \\ &= 7 \cdot (99 + 1) + 6 \cdot (9 + 1) + 5 \\ &= 7 \cdot 99 + 7 + 6 \cdot 9 + 6 + 5 \\ &= 7 \cdot 99 + 6 \cdot 9 + (7 + 6 + 5). \end{aligned}$$

  - Is 8514 divisible by 9?

  - Is 59814 divisible by 9?

DIVISIBILITY

- **Divisible by 9**

If the sum of the digits of an integer is a multiple of 9, then the integer is divisible by 9. Otherwise, the integer is not divisible by 9.

– Why?

– Example

  - 79637
  - 87653

DIVISIBILITY

- **Example**
  - Is 765 divisible by 3?

$$765 = 7 \cdot 99 + 6 \cdot 9 + (7 + 6 + 5).$$

  - Is 67242 divisible by 3?

  - Is 6148 divisible by 3?

DIVISIBILITY

- **Divisible by 3**

If the sum of the digits of an integer is a multiple of 3, then the integer is divisible by 3. Otherwise, the integer is not divisible by 3.

– Why?

– Example

  - 81645
  - 913529

DIVISIBILITY

- **Example**
  - *A* is the units digit in the four-digit number 463*A*. If 463*A* is divisible by 3 and by 4, then what are all the possible values of *A*?

DIVISIBILITY

- **Example**
  - N is the tens digit in the five-digit number 24,6N8. If 24,6N8 is divisible by 9, then must 24,6N8 divisible by 4?

DIVISIBILITY

- **Divisibility summary**

Number	Condition under which n is divisible by the number
2	Units digit of n is 0, 2, 4, 6, or 8
3	Sum of the digits of n is a multiple of 3
4	Number formed by last two digits of n is a multiple of 4
5	Units digit of n is 0 or 5
9	Sum of the digits of n is a multiple of 9
10	Units digit of n is 0

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PRIME NUMBERS

- **Definition**
  - A **prime number** is a positive integer that is divisible by exactly two positive integers: 1 and the number itself.
    - Examples: 2, 3, 5, 7, 11, 13, 17, 19, …
  - A **compositive number** is a positive integer that is divisible by more than two positive integers.
    - Examples: 4, 6, 8, 9, 12, 14, 15, ….
  - **The number 1 is NEITHER a prime NOR a composite number.**

PRIME NUMBERS

- **Example**
  - List every prime number that is less than 20.

PRIME NUMBER

- **Example**
  - Is 113 a prime number?
    - Divisible by 2?
    - Divisible by 3?
    - Divisible by 5?
    - Divisible by 7?
    - Divisible by 4, 6, 8, 9, 10?
    - Given that $11^2 = 121,$ do we need to test the numbers bigger than 10?

PRIME NUMBER

- **How to test whether a number is a prime number?**

1. To check if a number is prime, **we only need to test if that number is divisible by prime numbers**.

2. To check if a number is prime, **we only need to test if it is divisible by primes whose squares are less than or equal to the number we are testing.**

  - If a number is not divisible by a prime number, then it is not divisible by any multiple of that prime.

PRIME NUMBER

- **Example**
  - Is each of the following numbers prime or composite?
    - 61
    - 91
    - 143
    - 157

PRIME NUMBER

- **Example**
  - Find the largest two-digit **composite** number in which both digits are prime.

PRIME NUMBER

- **Example**
  - Find all pairs of prime whose sum is 61.

PRIME NUMBER

- **Example (Sieve of Eratosthenes)**
  - Start with the number grid, and perform the following process
    - 1. Circle the smallest number that is not circled or crossed out
    - 2. Cross out all multiples circled out in step 1.
    - 3. If every number is circled or crossed out, then stop. Otherwise go to step 1.

	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

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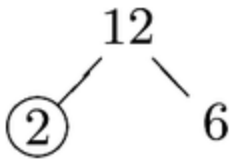
PRIME FACTORIZATION

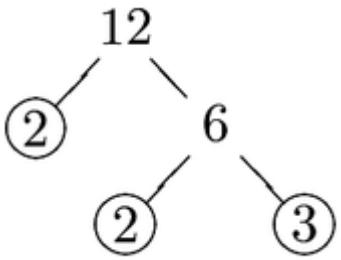
- **Prime factorization**
  - Express a number as the product of prime numbers.
  - Example

$$12 = 2 \cdot 6 = 2 \cdot 2 \cdot 3 = 2^2 \cdot 3^1$$

PRIME FACTORIZATION

- **Factor tree**
  - Example: find the prime factorization of 12





PRIME FACTORIZATION

- **Example**
  - Find the prime factorization of each of the following numbers:
    - 30, 60, 252, 288

PRIME FACTORIZATION

- **Fundamental Theorem of Arithmetic**

> Every positive integer has a **unique** prime factorization.

  - Prime numbers can be considered as building blocks of integers.

PRIME FACTORIZATION

- **Example**
  - Find the prime factorization of the following perfect squares.
    - 16, 36, 81, 144

PRIME FACTORIZATION

- **Prime factorization of perfect squares**

1. All of the **exponents** in the prime factorization of a perfect square are **even**.

2. Any prime factorization in which all of the **exponents are even** is the prime factorization of a perfect square.

PRIME FACTORIZATION

- **Example**
  - What positive integer squared equals 96 x 486?

PRIME FACTORIZATION

- **Example**
  - Paul multiplies two positive integers and gets a product of 16000. If neither of Paul's integers ends in 0, then what is the sum of Paul's integers?

PRIME FACTORIZATION

- **Example**
  - Let N be the product of the 10 smallest prime numbers. Explain why the prime factorization of $N+1$ must include a prime that isn't among the 10 smallest prime numbers.
  - Let N be the product of the 100 smallest prime numbers. Explain why the prime factorization of $N+1$ must include a prime that isn't among the 100 smallest prime numbers.
  - Explain why there is not a largest prime number. That is, explain why there are infinitely many primes.

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LEAST COMMON MULTIPLE

- **Common Multiple**
  - Suppose *a* and *b* are positive integers. If a number is a multiple of both *a* and *b*, then that number is a **common multiple** of *a* and *b*.
  - Example:
    - 6 is a multiple of 2 and 3.
    - 12 is a multiple of 2 and 3.
    - 18 is a multiple of 2 and 3.
- **Least Common Multiple (LCM)**
  - The smallest positive integer that is a multiple of both *a* and *b* is called the **least common multiple** of *a* and *b*.
  - Example
    - 6 is the least common multiple of 2 and 3.
    - 6 = lcm[2, 3]

LEAST COMMON MULTIPLE

- **Example**
  - List the five smallest multiple of 18.
  - List the five smallest multiple of 24.
  - What is the least common multiple of 18 and 24?

LEAST COMMON MULTIPLE

- **Example**
  - Find the prime factorizations of the five smallest positive multiples of 18. Compare these prime factorizations to the prime factorization of 18. Do you notice anything interesting?

LEAST COMMON MULTIPLE

- **Example**

The prime factorization of 72 is $2^3 \cdot 3^2$. Use the given prime factorization of each number below to determine if the number is a multiple of 72.

- $864 = 2^5 \cdot 3^3$.
- $400 = 2^4 \cdot 5^2$.
- $1008 = 2^4 \cdot 3^2 \cdot 7^1$.
- $2916 = 2^2 \cdot 3^6$.

LEAST COMMON MULTIPLE

- **Prime factorization of multiples**

> Let *n* be a positive integer. The prime factorization of any multiple of *n* includes the prime factorization of *n*.

LEAST COMMON MULTIPLE

- **Example**
  - Use the prime factorization of 24 and 90 to find lcm[24, 90].

LEAST COMMON MULTIPLE

- **Use prime factorization to find the least common multiple**

1. Find the prime factorization of each number.

2. The prime factorization of the LCM is the product of the **highest power of each prime factor** of all the numbers.

LEAST COMMON MULTIPLE

- **Example**
  - Find the least common multiples of the following numbers.
    - $\text{lcm}[96, 144]$.
    - $\text{lcm}[28, 35]$.
    - $\text{lcm}[2^2 \cdot 3^3 \cdot 5^1, 3^3 \cdot 5^2 \cdot 7^1]$.
    - $\text{lcm}[24, 36, 42]$.

LEAST COMMON MULTIPLE

- **Multiple of the least common multiple**

Let a and b be positive integers. Every multiple of $\text{lcm}[a, b]$ is a common multiple of a and b, and every common multiple of a and b is a multiple of $\text{lcm}[a, b]$

LEAST COMMON MULTIPLE

- **Example**
  - Compute $\text{lcm}[2 \cdot 500, 2 \cdot 300]$.
  - Compute $2\ \text{lcm}[500, 300]$.
  - Are they the same?

LEAST COMMON MULTIPLE

- **Multiple of the least common multiple**

For any positive integers a, b, and n, we have

$$\text{lcm}[na, nb] = n\text{lcm}[a, b].$$

LEAST COMMON MULTIPLE

- **Example**

  Compute $\text{lcm}[606060, 707070]$.

LEAST COMMON MULTIPLE

- **Example**
  - A church rings its bells every 15 minutes, the school rings its bells every 20 minutes, and the day care center rings its bells every 25 minutes. If they all ring their bells at noon on the same day, at what time will they next all ring their bells together?

LEAST COMMON MULTIPLE

- **Example**
  - The number 16128 is a multiple of 6, 7, and 8. What is the smallest multiple of 6, 7, and 8 that is greater than 16128?

LEAST COMMON MULTIPLE

- **Example**
  - What is the smallest positive integer greater than 1 that leaves a remainder of 1 when divided by each of 6, 7, and 8?

LEAST COMMON MULTIPLE

- **Example**
  - Find the smallest positive number that leaves a remainder of 5 when divided by 6, a remainder of 6 when divided by 7, and a remainder of 7 when divided by 8.

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DIVISORS

- **Definition**
  - Let a be a nonzero integer and b an integer.

We say that a is a divisor of b if b is a multiple of a.

- Example
  - 7 is a multiple of 21 because $21 = 7 \cdot 3$.
  - 8 is NOT a divisor of 21.

DIVISORS

- **Divisor, factor, multiple**
  - If a and b are nonzero integers, then all the following statements are the same:
    - *a* is a **divisor** of *b*.
    - *a* is a **factor** of *b*.
    - *b* is a **multiple** of *a*.
    - *b* is **divisible** by *a*.

DIVISORS

- **Example**
  - Find all the positive divisors of 84.

DIVISORS

- **Example**
  - Count the positive divisors of each integer from 8 to 18. Which of these integers have an odd number of divisors?

Number	Divisors	Count	Number	Divisors	Count
8	1, 2, 4, 8	4	14	1, 2, 7, 14	4
9	1, 3, 9	3	15	1, 3, 5, 15	4
10	1, 2, 5, 10	4	16	1, 2, 4, 8, 16	5
11	1, 11	2	17	1, 17	2
12	1, 2, 3, 4, 6, 12	6	18	1, 2, 3, 6, 9, 18	6
13	1, 13	2			

DIVISORS

- **Number of divisors of perfect squares**

- A perfect squre has an odd number of divisors.
- If a number has an odd number of divisors, then it is a perfect square.

DIVISORS

- **Example**
  - Note that 20 is a divisor of 60.
    - Must every divisor of 20 also be a divisor of 60?
    - Must every divisor of 60 also be a divisor of 20?

DIVISORS

- **Number of divisors of perfect squares**

> If a is a divisor of b, then every divisor of a is also a divisor of b.

  - Why?

DIVISORS

- **Number of divisors of perfect squares**

Let a, b, and c be integers. If c is a divisor of a and of b, then c is a divisor of $a + b$ and of $a - b$.

  - Why?

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GREATEST COMMON DIVISOR

- **Common Divisor**
  - If a number is a divisor of both *a* and *b*, then that number is a **common divisor** of *a* and *b*. (Also called as the **common factor** of a and b.)
  - Example:
    - 2 is a common divisor of 18 and 24.
    - 3 is a common divisor of 18 and 24.
    - 6 is a common divisor of 18 and 24.
- **Greatest Common Divisor (GCD)**
  - The greatest positive integer that is a divisor of both *a* and *b* is called the **greatest common divisor** of *a* and *b*.
  - Example
    - 6 is the greatest common divisor of 18 and 24.
    - 6 = gcd[18, 24]

GREATEST COMMON DIVISOR

- **Example**
  - Find the prime factorization of 12.
  - Find the prime factorization of each positive divisors of 12.

  - The prime factorization of a number includes the prime factorization of each of its divisors.

GREATEST COMMON DIVISOR

- **Example**

The prime factorization of 1944 is $2^3 \cdot 3^5$. Below are three numbers and their prime factorizations. In each part, use the prime factorization of the number to determine if the number is a divisor of 1944.

- (a) $108 = 2^2 \cdot 3^3$.
- (b) $45 = 3^2 \cdot 5^1$.
- (c) $48 = 2^4 \cdot 3^1$.

GREATEST COMMON DIVISOR

- **Number of divisors of perfect squares**

Let n be a positive integer. The prime factorization of n includes the prime factorization of each divisor of n.

– Example

The prime factorization of 24 is $2^3 \times 3^1$.
The prime factorization of 6 is $2^1 \cdot 3^1$.
The prime factorization of 8 is 2^3.

GREATEST COMMON DIVISOR

- **Example**
  - Use the prime factorizations of 360 and 48 to find gcd[360, 48].

GREATEST COMMON DIVISOR

- **Example**
  - Use the prime factorizations of 144 and 64 to find gcd[144, 64].

GREATEST COMMON DIVISOR

- **Relatively prime**

If two integers do not have any prime factors in common, then the integers are **relatively prime**.

- If two integers are relatively prime, then their GCD is 1.

- Example
  - 2 and 7 are relatively prime.
  - 5 and 13 are relatively prime.
  - 10 and 21 are relatively prime.

GREATEST COMMON DIVISOR

- **Example**
  - Are the following pairs relatively prime?
    - 5 and 13
    - 6 and 13
    - 6 and 35
    - 21 and 49

GREATEST COMMON DIVISOR

- **Example**
  - Compute each of the following
    - (a) $\gcd(27, 39)$.
    - (b) $\gcd(100, 63)$.
    - (c) $\gcd(2^3 \cdot 5^3 \cdot 11^2, 3^2 \cdot 5^2 \cdot 11^1)$.
    - (d) $\gcd(72, 240, 288)$.

GREATEST COMMON DIVISOR

- **GCD and divisors**

Let a and b be positive integers. Every common divisor of a and b is a divisor of $\gcd(a, b)$, and every divisor of $\gcd(a, b)$ is a common divisor of a and b.

– Example

The GCD of 16 and 24 is $\gcd(16, 24) = 8$.

The common divisors of 16 and 24 are: 1, 2, 4, 8.

The common divisors of 16 and 24 are divisors of $\gcd(16, 24)$.

Every divisor of $\gcd(16, 24)$ is a common divisor of 16 and 24.

GREATEST COMMON DIVISOR

- **Example**
  - (a) Are $\gcd(2 \cdot 18, 2 \cdot 24)$ and $2\gcd(18, 24)$ equal?.
  - (b) Do we have $\gcd(2a, 2b) = 2\gcd(a, b)$?

GREATEST COMMON DIVISOR

- **Multiples of GCD**

> For any positive integers a, b, and n, we have
>
> $$\gcd(na, nb) = n \gcd(a, b).$$

– Example

Compute $\gcd(606060, 707070)$.

GREATEST COMMON DIVISOR

- **Example**
  - Every bag of candy in the Grab-bag Candy store has the same number of candies. Tony and Kaya each grab some bags of candies. Tony gets a total of 70 candies and Kaya gets a total of 42 candies. What is the smallest possible number of bags Tony could have grabbed?

GREATEST COMMON DIVISOR

- **Example**
  - (a) Find $\gcd(4, 9)$ and $\text{lcm}[4, 9]$.
  - (b) Find $\gcd(10, 27)$ and $\text{lcm}[10, 27]$.

For any positive integers a, b that are relatively prime, we have

$$\text{lcm}(a, b) = ab.$$

GREATEST COMMON DIVISOR

- **Example**
  - (a) Must a number that is a multiple of 4 and of 9 be a multiple of 36?
  - (b) Must a number that is a multiple of 3 and of 12 be a multiple of 36?
  - (c) Find the digit A such that $59{,}7A6$ is a multiple of 36.

To test for divisibility by a composite number, we can test for divisibility by two relatively prime numbers whose product is the composite number.

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SUMMARY

- **Prime number**
  - A **prime number** is a positive integer that is divisible by exactly two positive integers: 1 and the number itself.

- **Multiples and divisors**
  - Let *a* and *b* be integers. We say that *a* is a **multiple** of *b* if *a* equals *b* times some integer.
  - If *b* is nonzero and *a* is a multiple of *b*, then *b* is a **divisor** of *a*.
  - If *a* and *b* are multiples of *c*, then *a*+*b* and *a*-*b* are multiples of c.
  - If *a* is a multiple of *b*, then every multiple of *a* is also a multiple of *b*.

- **Least common multiple (LCM) and greatest common divisor (GCD)**

$$\text{lcm}[na, nb] = n\text{lcm}[a, b].$$

$$\gcd(na, nb) = n \gcd(a, b).$$